

GOVT 6019: Homework 4 — Solutions

Fall 2026 · Expectation

Melissa Sands (Instructor)

Izza Malik (Teaching Assistant)

Question 1: Name the distribution

(i) $X \sim \mathcal{U}\{1, \dots, 7\}$, discrete uniform on the seven ballot positions. Every ordering is equally likely by statute, so every position is equally likely for any one candidate, and $f(x) = 1/7$ at each.

(ii) $X \sim \text{Bernoulli}(p)$, with p the share of the district supporting the incumbent. One trial, two outcomes, coded 1 and 0.

(iii) $X \sim \text{Binomial}(15, 0.4)$. Fifteen independent trials, each with the same success probability .4, and X counts the successes.

Independence is doing real work in this one. A survey that sampled precincts first and then interviewed several voters within each selected precinct — cluster sampling — would draw respondents who share a neighborhood, and neighbors resemble one another politically. The fifteen responses would then carry less information than fifteen independent draws, and the Binomial would understate how much the count varies from sample to sample.

(iv) $X \sim \text{Poisson}(1.5)$. A count of events in a fixed window — one year — arriving at an average rate $\lambda = 1.5$ per window.

(a) $\mathbb{P}(X = 1) = 1/7 = .143$. The top two positions are disjoint events, so $\mathbb{P}(X \leq 2) = 1/7 + 1/7 = 2/7 = .286$.

(b) Writing S for a supporter and N for a non-supporter, the sequences of four responses with exactly two S 's are

$SSNN, SNSN, SNNS, NSSN, NSNS, NNSS,$

six of them, which matches $\binom{4}{2} = \frac{4!}{2!2!} = \frac{24}{4} = 6$. Each of the six has probability $(.4)^2(.6)^2 = .16 \times .36 = .0576$, and they are disjoint, so

$$\mathbb{P}(X = 2) = 6 \times .0576 = .3456.$$

The coefficient counts the sequences; the rest is the probability of any one of them.

(c)

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dbinom(6, size = 15, prob = 0.4)      # P(X = 6)
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[1] 0.2065976
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pbinom(5, size = 15, prob = 0.4) # P(X <= 5)
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[1] 0.4032156
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$\mathbb{P}(X = 6) = .207$ and $\mathbb{P}(X \leq 5) = .403$. By hand the first is $\binom{15}{6}(.4)^6(.6)^9 = 5005 \times .004096 \times .010078$.

Question 2: Two ways to reach the hospital

(a) Helicopter .48 among serious accidents and .16 among less serious ones. Ambulance .60 and .20. The helicopter death rate is lower in both severity groups.

(b) Helicopter $64/200 = .32$. Ambulance $260/1100 = .2364$. The ambulance death rate is lower overall.

(c)

$$\text{helicopter : } .48 \left(\frac{100}{200} \right) + .16 \left(\frac{100}{200} \right) = .32$$

$$\text{ambulance : } .60 \left(\frac{100}{1100} \right) + .20 \left(\frac{1000}{1100} \right) = .2364$$

Each weight is the proportion of that mode's own patients falling in that severity group. Half of the helicopter patients are serious cases, against $100/1100 = .09$ of the ambulance patients.

(d) Severity is assessed at the scene, before transport is chosen, so the helicopter is dispatched to the serious accidents. Which mode a patient receives depends on X . That dependence is what makes the weights differ between the two rows.

(e) Two groups of patients differ in their pooled rate either because the group means $\bar{Y}_{X=x}$ differ, or because the weights $\hat{p}(x)$ differ. Here the group means favor the helicopter in both severity groups. Only the distribution of severity differs.

(f) The pooled comparison answers: what was the death rate among all helicopter patients, against all ambulance patients. A reader takes it to answer: what would happen to a given patient transported one way rather than the other. The two coincide only when severity is distributed the same way across the modes.

(g) Simpson's paradox. In Week 1 the relationship between income and Republican vote ran one way within states and the other way across states. In both cases a comparison reverses when a grouping variable is collapsed, because the groups enter the pooled figure with different weights. There X is the state; here X is severity.

Question 3: The utility of voting

(a) $U_{\text{vote}} = B - C = 495$ with probability p , and $-C = -5$ with probability $1 - p$. $U_{\text{abstain}} = 0$ with probability 1.

(b)

$$\mathbb{E}[U_{\text{vote}}] = p(B - C) + (1 - p)(-C) = pB - C = .01 - 5 = -4.99, \quad \mathbb{E}[U_{\text{abstain}}] = 0.$$

The model predicts abstaining. Reaching $pB - C$ directly is the same calculation: the $-C$ is paid on both branches, so it comes out of the expectation by linearity.

(c) $\mathbb{E}[U_{\text{vote}}] = pB - C$ and $\mathbb{E}[U_{\text{abstain}}] = 0$. The model predicts voting when $pB > C$.

(d) Setting $pB = C$ gives $p^* = C/B = 5/500 = .01$. The citizen is indifferent only if one election in a hundred is decided by a single vote. At $p = 10^{-7}$ the expected benefit is $pB = 5 \times 10^{-5}$, five orders of magnitude below the cost. The model predicts that nobody votes.

(e) Voting requires $pB - C + D > 0$, that is $D > C - pB$. With these numbers $D > 4.99$. The cost is paid with certainty and the benefit almost never arrives, so D carries nearly the whole decision.

(f) The four quantities differ across citizens, so a model of this form predicts variation in turnout across a population rather than one universal choice.

(g) A constant a is a random variable that takes one value, so the third property applies to the sum $a + bU$:

$$\begin{aligned}\mathbb{E}[a + bU] &= \mathbb{E}[a] + \mathbb{E}[bU] \quad (\text{third property: } \mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]) \\ &= a + \mathbb{E}[bU] \quad (\text{first property: } \mathbb{E}[c] = c) \\ &= a + b\mathbb{E}[U] \quad (\text{second property: } \mathbb{E}[cX] = c\mathbb{E}[X])\end{aligned}$$

(h) Applying the new scale to both actions,

$$\mathbb{E}[a + bU_{\text{vote}}] - \mathbb{E}[a + bU_{\text{abstain}}] = b(\mathbb{E}[U_{\text{vote}}] - \mathbb{E}[U_{\text{abstain}}]).$$

The a cancels because it is added to both, and $b > 0$ leaves the sign of the difference unchanged, so the predicted action is the same. A constant added to the utility of voting alone does not cancel. It shifts the comparison by exactly that amount, and it is D in (e).

Question 4: Unseen vignettes

(a) Version j goes unseen when each of the 20 respondents lands on one of the other nine versions. Each respondent does so with probability .9, independently of the others, so

$$\mathbb{P}(I_j = 1) = (.9)^{20} = .122.$$

I_j is a Bernoulli variable, so $\mathbb{E}[I_j] = \mathbb{P}(I_j = 1) = .122$. This is the general identity $\mathbb{P}(A) = \mathbb{E}[I_A]$ for any event A .

(b) $U = I_1 + I_2 + \cdots + I_{10}$. By linearity,

$$\mathbb{E}[U] = \sum_{j=1}^{10} \mathbb{E}[I_j] = 10 (.9)^{20} = 1.22.$$

On average, about one version in ten gets no respondents.

(c) No. If version 1 goes unseen, all 20 respondents fall among the other nine versions. Version 2 then catches each respondent with probability $1/9$ rather than $1/10$, so $\mathbb{P}(I_2 = 1 \mid I_1 = 1) = (8/9)^{20} = .095$,

below the unconditional .122. Linearity of expectation requires no independence, so the sum in (b) is correct regardless.

(d) There are $\binom{20}{2} = 190$ pairs of respondents. A given pair sees the same version with probability $1/10$: whatever the first respondent draws, the second matches it with probability $1/10$. By linearity the expected number of matching pairs is $190 \times 1/10 = 19$. This is the birthday problem with 10 “days” in place of 365.

(e)

```
set.seed(6019)
U <- replicate(10000, 10 - length(unique(sample(1:10, 20, replace = TRUE))))
mean(U)           # compare with 10 * 0.9^20 = 1.216
```

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[1] 1.218
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```
mean(U > 0)       # share of experiments with at least one unseen version
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[1] 0.7869
```

The simulated mean of U is close to 1.22. At least one version goes unseen in about 79 percent of the simulated experiments.

The expected count in (b) is small. The probability of a failed design is not. With 20 respondents and 10 versions, simple randomization, which assigns each respondent independently, leaves some version with no respondents roughly four times in five. The probability falls as the sample grows: about .05 at $n = 50$ and about .0003 at $n = 100$. Small samples are where simple randomization most often fails.

The remedy is to fix the group sizes in advance. Complete randomization shuffles a list containing each version exactly twice and deals it out to the 20 respondents. Every version is then seen by exactly two respondents. Block randomization applies the same step within subgroups of respondents.

The exact probability that at least one version goes unseen requires inclusion–exclusion over all subsets of versions. The expected number of unseen versions in (b) required only $\mathbb{P}(I_j = 1)$ and addition. Indicators turn a hard probability problem into an easy expectation problem.

Question 5: Contact and turnout

The marginal distributions come from the row and column sums: $\mathbb{P}(C = 1) = .05 + .25 = .30$ and $\mathbb{P}(T = 1) = .40 + .25 = .65$.

(a) C is an indicator, so $\mathbb{E}[C] = \mathbb{P}(C = 1) = .30$. From the definition,

$$\mathbb{V}[C] = .7(0 - .3)^2 + .3(1 - .3)^2 = .063 + .147 = .21.$$

The Bernoulli variance gives the same value: $p(1 - p) = .3 \times .7 = .21$.

(b) T takes only the values 0 and 1, and $0^2 = 0$ and $1^2 = 1$. So $T^2 = T$ for every outcome, and $\mathbb{E}[T^2] = \mathbb{E}[T] = .65$. Then

$$\mathbb{V}[T] = .65 - .65^2 = .65 - .4225 = .2275.$$

(c) Write $\mu_C = \mathbb{E}[C]$ and $\mu_T = \mathbb{E}[T]$. Both are constants.

$$\begin{aligned}\text{Cov}(C, T) &= \mathbb{E}[(C - \mu_C)(T - \mu_T)] \\ &= \mathbb{E}[CT - \mu_T C - \mu_C T + \mu_C \mu_T] && \text{(expand the product)} \\ &= \mathbb{E}[CT] - \mu_T \mathbb{E}[C] - \mu_C \mathbb{E}[T] + \mu_C \mu_T && \text{(linearity)} \\ &= \mathbb{E}[CT] - \mu_C \mu_T - \mu_C \mu_T + \mu_C \mu_T \\ &= \mathbb{E}[CT] - \mathbb{E}[C] \mathbb{E}[T].\end{aligned}$$

$CT = 1$ only when both indicators equal 1, so $\mathbb{E}[CT] = \mathbb{P}(C = 1, T = 1) = .25$. Then

$$\text{Cov}(C, T) = .25 - .30 \times .65 = .25 - .195 = .055.$$

The covariance is positive: contacted voters turn out more often. Campaigns target voters who are already likely to turn out, such as past voters and registered partisans. That targeting alone produces a positive covariance, with or without any effect of contact. The covariance describes the joint distribution. It does not say which process produced it.

(d) By the variance of a sum,

$$\mathbb{V}[K] = \mathbb{V}[C] + \mathbb{V}[T] + 2 \text{Cov}(C, T) = .21 + .2275 + 2(.055) = .5475.$$

Directly, $K = 0$ when neither event occurs, $K = 1$ when exactly one occurs, and $K = 2$ when both occur:

k	0	1	2
$\mathbb{P}(K = k)$	.30	.40 + .05 = .45	.25

$\mathbb{E}[K] = .45 + 2(.25) = .95$ and $\mathbb{E}[K^2] = .45 + 4(.25) = 1.45$, so $\mathbb{V}[K] = 1.45 - .95^2 = 1.45 - .9025 = .5475$. The two methods agree.

Treating C and T as independent drops the covariance term and gives $.21 + .2275 = .4375$. That understates the variance of K by .11.

(e) $\mathbb{E}[X] = \frac{1}{3}(-1 + 0 + 1) = 0$. $XY = X^3$, which takes the values $-1, 0$ and 1 with probability $1/3$ each, so $\mathbb{E}[XY] = 0$. Then

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y] = 0 - 0 \times \mathbb{E}[Y] = 0.$$

$Y = 0$ exactly when $X = 0$. So $\mathbb{P}(X = 0, Y = 0) = \mathbb{P}(X = 0) = 1/3$. But $\mathbb{P}(Y = 0) = 1/3$ as well, so $\mathbb{P}(X = 0) \mathbb{P}(Y = 0) = 1/9 \neq 1/3$. X and Y are not independent: Y is a function of X . Covariance measures only linear association, and the relationship here is symmetric about zero, so the linear association is zero.